



AXP313A PMIC For Multi-Core High-Performance System

1. Characteristics

• 3 DCDCs

DCDC1: 0.5~1.2V, 10mV/step, 1.22~1.54V,
20mV/step, 1.6~3.4V, 0. 1V/step,
IMAX=1.5A

DCDC2: 0.5~1.2V, 10mV/step, 1.22~1.54V,
20mV/step, IMAX=3A@VIN=5V,

IMAX=2.5A@VIN=3.7V

DCDC3: 0.5~1.2V, 10mV/step, 1.22~1.84V,
20mV/step, IMAX=2A

• 3 LDOs

RTCLDO: 1.8V/2.5V/2.8V/3.3V,
IMAX=30mA

ALDO1: 0.5~3.5V, 0.1V/step
IMAX=300mA

DLDO1: 0.5~3.5V, 0.1V/step
IMAX=500mA

Supports TWIS (Two Wire Serial Interface) communication.

In signal mode, the slave address can be configured to 0x36 or...
0x37 (7 bits)

Internal temperature sensor and over-temperature protection

• Can monitor DC-DC output voltage, transmit interrupts, undervoltage/overvoltage.

Overvoltage protection

The startup timing and default voltage of a DC-DC/LDO can be determined.

system

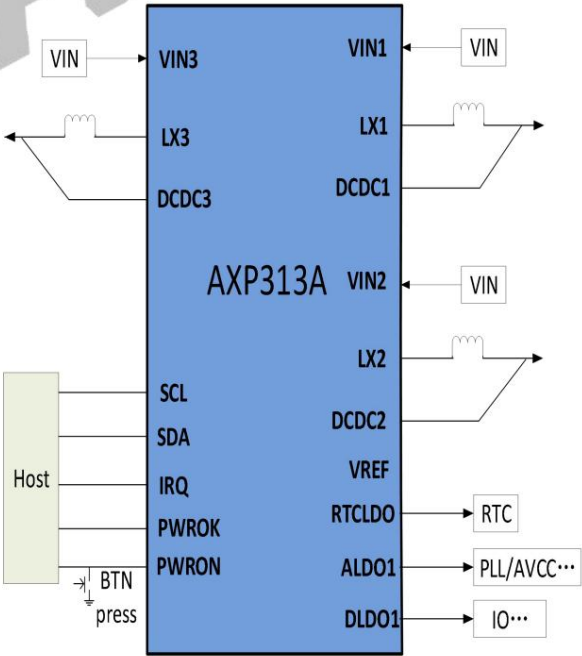
The AXP313A supports 6 power outputs (including 3 DC-DC converters).
(And 3 LDOs). To ensure the safety and stability of the power system,
The AXP313A integrates overvoltage protection (OVP) and undervoltage protection (UVP).
Overcurrent (OCP) and overtemperature (OTP) protection circuits.
In addition, the AXP313A features power on/off, sleep/wake-up, and other electrical functions.
The source management function reflects the management value of chip integration.

The AXP313A supports TWIS, allowing the system to dynamically adjust the output.
Output voltage, control power supply output and shutdown, flexible configuration
Interrupt management and sleep/wake conditions.

Chip Information

model	Packaging	size
AXP313A	QFN-20	3mm * 3mm

Simplified application diagram



2 Applications

• OTT Box

• IPC

3 Overview

The AXP313A is a highly integrated power management chip.
For applications requiring multiple power conversion outputs, it can be used with
Other BMUs, in conjunction with power solutions for battery application scenarios, constitute the power supply solution.
This solution can also be used in low-power devices such as IPCs and smart speakers.
Battery solutions fully meet the power requirements of application processor systems.
The source requires relatively complex and precise control.



Version History

Version	Date	Responsible Person	Version Description
1.0	2021-05-31	Create a document	using AWA 1017.



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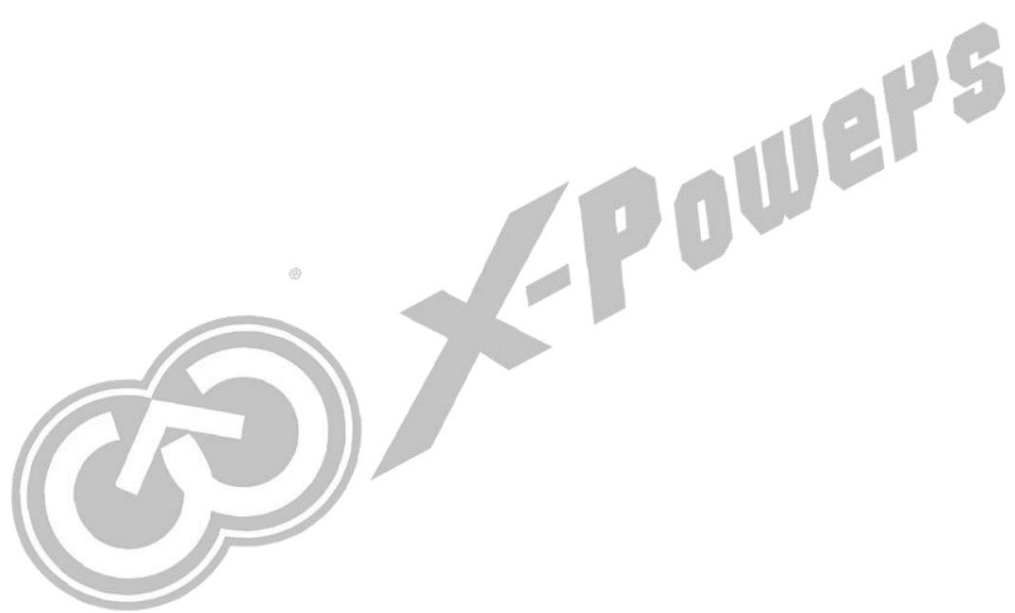


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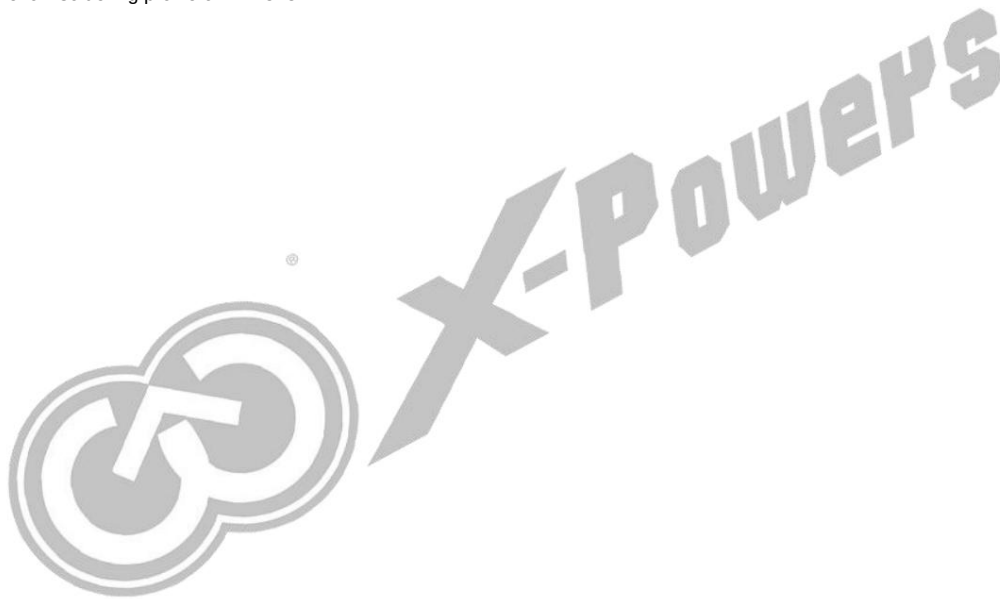




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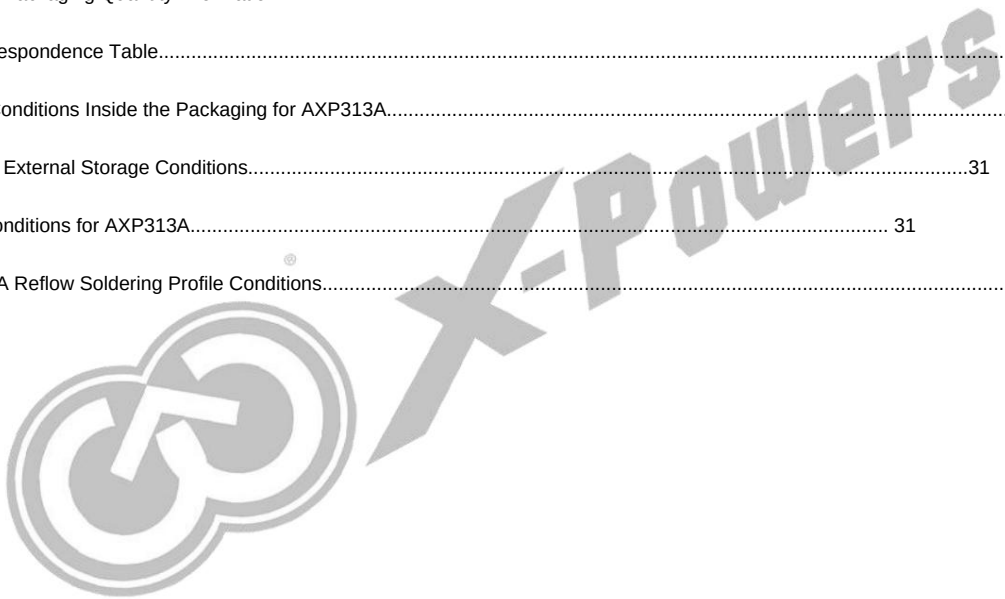
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4. Pin Configuration and Functions

Figure 4-1 Pin Map

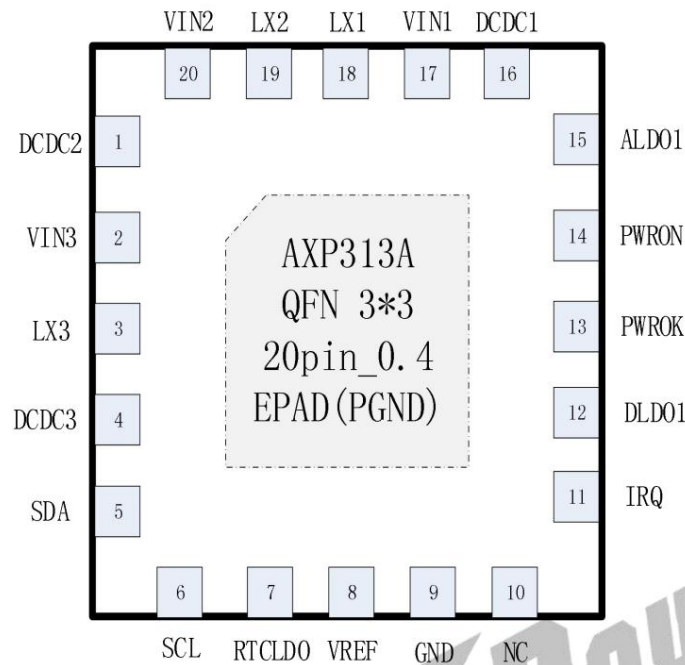


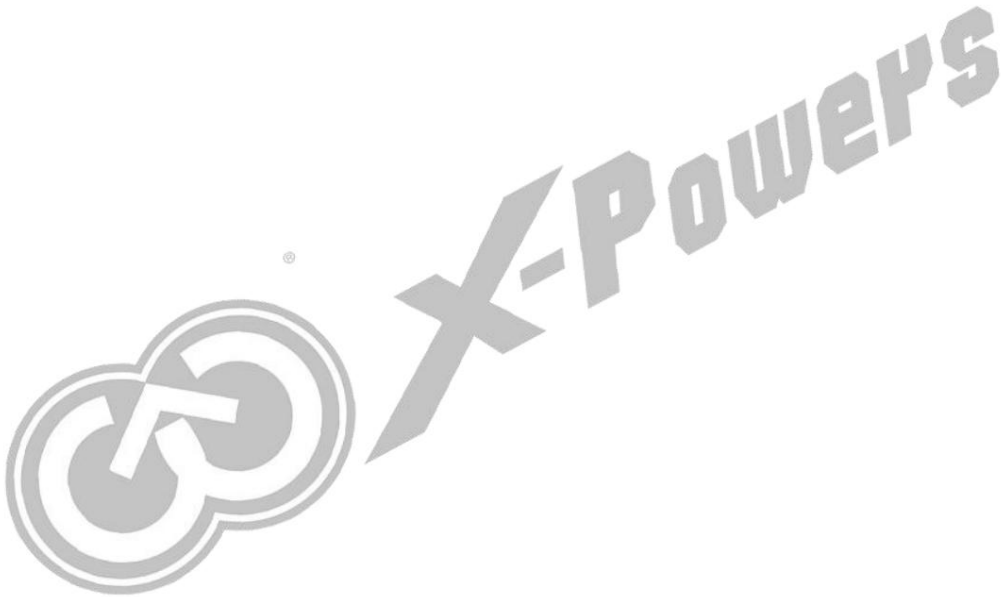
Table 4-1 Pin Description

pin		I/O(1)	describe
Serial Number	name		
1	DCDC2	I	DCDC2 feedback signal
2	VIN3	PI	DC-DC3 power input, external input capacitor
3	LX3	PO	DC-DC3 switching node, external inductor
4	DCDC3	I	DCDC3 feedback signal
5	SDA	IO	TWSI Data
6	SCL	IO	TWSI Clock Input
7	RTCLDO	PO	It can be customized as a normally open LDO for powering the RTC module.
8	VREF	AIO	Internal reference voltage, external 1uF capacitor.
9	GND	GND	Chip simulation
10	NC	/	Float
11	IRQ	IO	Interrupt indication/Power-on/Wake-up
12	DLDO1	PO	DLDO1 output, external output capacitor
13	PWROK	IO	Chip power on/off status indicator
14	PWRON	I	Power button, usually an external button
15	ALDO1	PO	ALDO1 output, external output capacitor



16	DCDC1	I	DCDC1 feedback signal
17	VIN1	PI	DC-DC1 power input, external input capacitor
18	LX1	PO	DCDC1 switching node, external inductor
19	LX2	PO	DC-DC2 switching node, external inductor
20	VIN2	PI	DC-DC2 power input, external input capacitor

(1) O represents output, I represents input, IO represents input/output, D represents digital, A represents analog, P represents power supply, and G represents ground.



5 Specifications

5.1 Limiting parameters (1)

Table 5-1 Limiting Parameters

SYMBOL	DESCRIPTION	MIN	MAX	UNIT
VIN1/VIN2/VIN3	Input Voltage	-0.3	7	V
Ta	Operating Temperature Range	-40	85	°C
TJ	Junction Temperature Range	-40	125	°C
Ts	Storage Temperature Range	-40	150	°C
TLEAD	Maximum Soldering Temperature (at leads, 10 seconds)		300	°C

(1) Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only. Functional operation of the device at these or any other conditions beyond those indicated under recommended operating conditions is not implied. Exposure to absolute maximum rated conditions for extended periods may affect device reliability.

5.2 ESD Level

Table 5-2 ESD Levels

		VALUE	UNIT
VESD	Human body model (HBM) (1)	±2000	V
	Charged device model (CDM) (2)	±750	V

(1) Reference: ESDA/JEDEC JS-001-2017.

(2) Reference: JEDEC EIA/JESD22-C101F.

5.3 Recommended Working Conditions

Table 5-3 Recommended Working Conditions

SYMBOL	DESCRIPTION	MIN	MAX	UNIT
VIN	Input voltage	3.0	5.5	V

5.4 Thermal resistance parameters

Table 5-4 Thermal Resistance Parameters

Thermal Metric(1)		VALUE	UNIT
θJA	Junction-to-ambient thermal resistance	32.6	°W
θJB	Junction-to-board thermal resistance	6.82	
θJC	Junction-to-case(top) thermal resistance	12.32	

(1) Thermal metrics are calculated refer to JEDEC document JESD51. The values are based on simulation.

5.5 Electrical Parameters

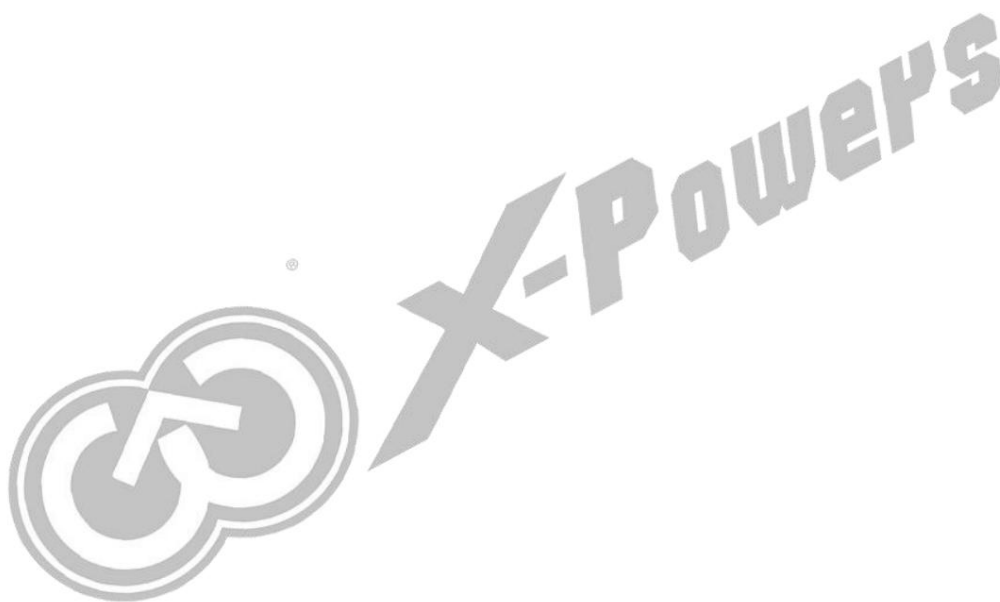
VIN = 5V, TA = -40°~85°

Table 5-5 Electrical Parameters

SYMBOL	DESCRIPTION	CONDITIONS	MIN	TYP	MAX	UNIT
PMIC Under Voltage						
VOFF	PMIC Under Voltage Power off			2.6		V
Off Mode Current						
IOFF	OFF Mode Current	VIN=5V		10		µA
TWSI						
VCC	Input Supply Voltage		1.8	3.3		V
ADDRESS	TWSI Slave Address (7 bits)			0x36/ 0x37		
fSCK	Clock Operating Frequency			0.4		MHz
VIL	SCK/SDA Logic Low Voltage				0.3V _{CC}	V
VIH	SCK/SDA Logic High Voltage		0.7V _{CC}			MHz
DCDC						
fOSC	Oscillator Frequency	Default		3		MHz
DCDC1						
VIN1	VIN1 Input Voltage		VOFF		5.5	V
IDC1OUT	Available Output Current			1500		mA
VDC1OUT	Output Voltage Range		0.5		3.4	V
VDC1_STEP	Output Voltage Step	VDC1OUT = 0.5~1.2V		10		mV/ step
		VDC1OUT = 1.22~1.54V		20		
		VDC1OUT = 1.6~3.4V		100		
VDC1_ACC	Output Voltage Accuracy	PWM Mode, VDC1OUT < 1V		±30		mV
		PWM Mode, VDC1OUT > 1V		±3%		
VDC1_OVP	Over Voltage Protection			120%* VDC1OUT		V
VDC1_UVP	Under Voltage Protection			85%* VDC1OUT		V

DCDC2						
VIN2	VIN2 Input Voltage		VOFF		5.5	V
IDC2OUT	Available Output Current	VIN=5V		3000		mA
		VIN=3.7V		2500		mA
VDC2OUT	Output Voltage Range		0.5		1.54 V	
VDC2_STEP	Output Voltage Step	VDC2OUT = 0.5~1.2V		10		mV/step
		VDC2OUT = 1.22~1.54V		20		mV/step
VDC2_ACC	Output Voltage Accuracy	PWM Mode, VDC2OUT < 1V		±30		mV
		PWM Mode, VDC2OUT > 1V		±3%		
VDC2_OVP	Over Voltage Protection			120%* VDC2OUT		V
VDC2_UVP	Under Voltage Protection			85%* VDC2OUT		V
DCDC3						
VIN3	VIN3 Input Voltage		VOFF		5.5	V
IDC3OUT	Available Output Current			2000		mA
VDC3OUT	Output Voltage Range		0.5		1.84 V	
VDC3_STEP	Output Voltage Step	VDC3OUT = 0.5~1.2V		10		mV/step
		VDC3OUT = 1.22~1.84V		20		mV/step
VDC3_ACC	Output Voltage Accuracy	PWM Mode, VDC3OUT < 1V		±30		mV
		PWM Mode, VDC3OUT > 1V		±3%		
VDC3_OVP	Over Voltage Protection			120%* VDC3OUT		V
VDC3_UVP	Under Voltage Protection			85%* VDC3OUT		V
RTCLDO						
VRTCLDO	Output Voltage	IRTC_VCC=1mA		1.8/ 2.5/ 2.8/ 3.3		V
IRTCCLDO	Output Current			30		mA
ALDO1						
VALDO1	Output Voltage Range	IALDO1=1mA	0.5		3.5	V
VALDO1_STEP	Output Voltage Step			100		mV/step
VALDO1_ACC	Output Voltage Accuracy	VALDO1<1V		±30		mV
		VALDO1>1V		±3%		
IALDO1	Output Current			300		mA
PSRR	Power Supply Rejection Ratio	Vin=3.7V, IALDO1=100mA, 1kHz		30		dB
eN	Output Noise, 0-80kHz Vout=1.8V			30		µVRMS

DLDO1						
VDLDO1	Output Voltage Range	IDLDO1=1mA	0.5		3.5	V
VDLDO1_STEP	Output Voltage Step			100		mV/ step
VDLDO1_ACC	Output Voltage Accuracy	VDLDO1<1V		±30		mV
		VDLDO1>1V		±3%		
IDLDO1	Output Current			500		mA
PSRR	Power Supply Rejection Ratio	Vin=3.7V, IDLDO1=100mA, 1kHz		30		dB
eN	Output Noise, 0-80kHz Vout=1.8V			55		µVRMS



6. Detailed Description

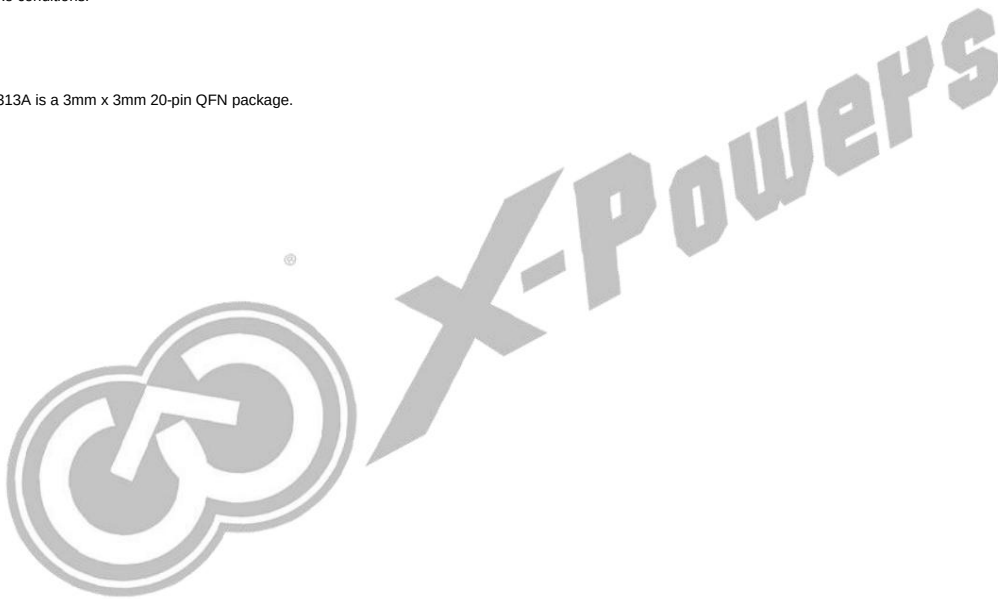
6.1 Overview

The AXP313A is a highly integrated power management chip designed for applications requiring multi-channel power conversion output. It can be used in conjunction with other BMUs to form power solutions for battery-powered applications, or in low-power, battery-free solutions such as IPCs and smart speakers, fully meeting the relatively complex and precise power control requirements of application processor systems.

The AXP313A supports six power outputs (including three DC-DC converters and three LDO converters). To ensure the safety and stability of the power system, the AXP313A integrates overvoltage (OVP), undervoltage (UVP), overcurrent (OCP), and overtemperature (OTP) protection circuits. In addition, the AXP313A features power management functions such as power-on/off and sleep/wake-up, demonstrating the value of integrated chip management.

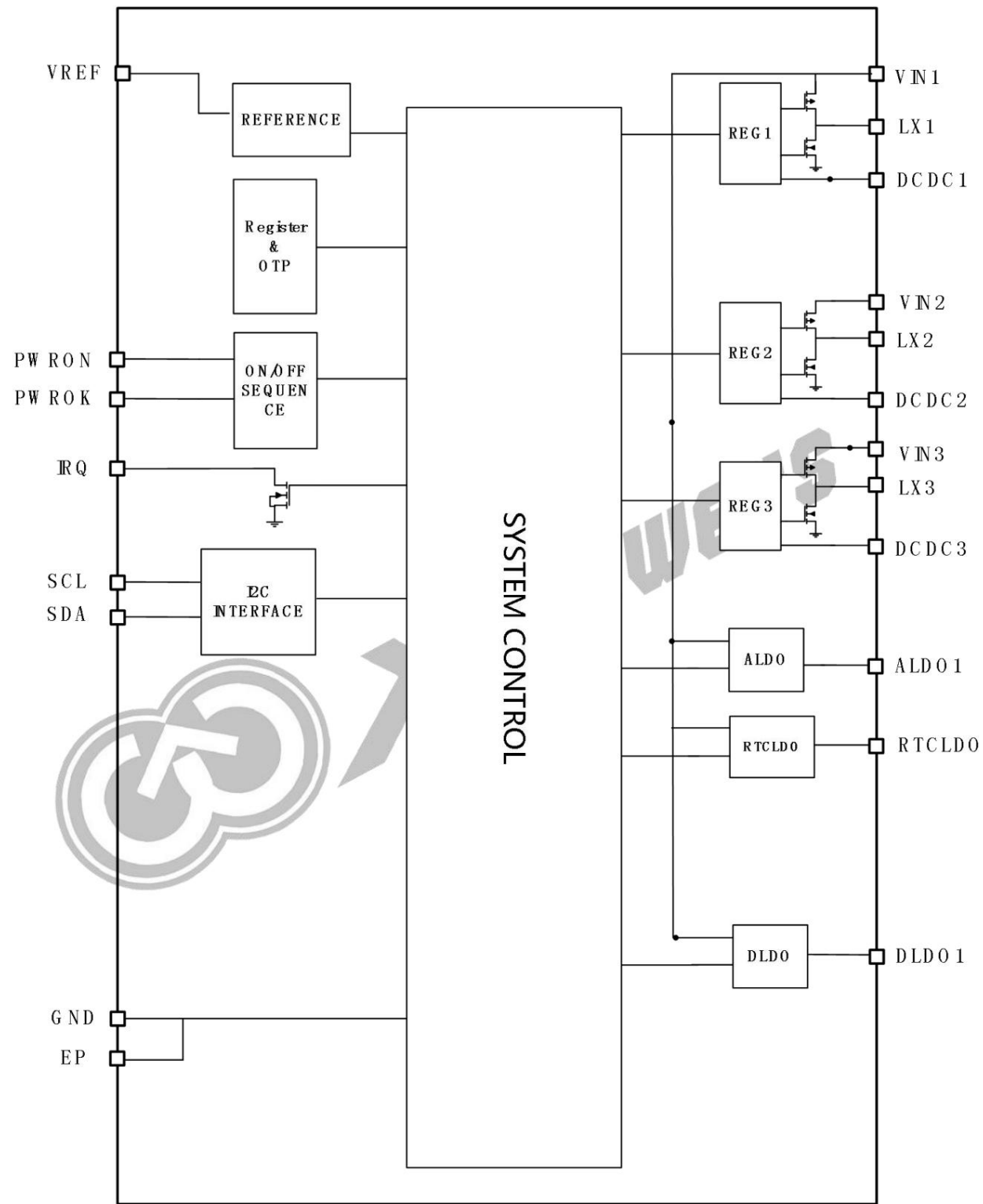
The AXP313A supports TWSI, allowing the system to dynamically adjust the output voltage, control power supply output and shutdown, and flexibly configure interrupt management and sleep/wake conditions.

The AXP313A is a 3mm x 3mm 20-pin QFN package.



6.2 Structural Diagram

Figure 6-1 Structural block diagram



6.3 TWI communication

When the AXP313A is working, the SCL/SDA pins of the TWI interface are pulled up to the system I/O power supply. The host can then use this interface to flexibly adjust and monitor the operating status of the AXP313A and obtain a wealth of information.

Note: "Host" refers to the main processor of the application system.

6.4 Operating Modes and Reset

The AXP313A has a power-off state and a power-on state. In the power-off state, all outputs except RTCLDO (if customized to be normally on) are turned off, and the total power consumption of the chip is about 5.5uA (about 10uA when customized to be normally on RTCLDO).

Power on/off key (POK)

The EN/PWRON pin is a function-multiplexed pin, customized by the customer, and is used as the PWRON pin by default. Connect a button between the PWRON pin and GND as an independent power on/off key (POK). The AXP313A can automatically recognize the "rising edge," "falling edge," "long press," and "short press" of this button and respond accordingly.

Power on

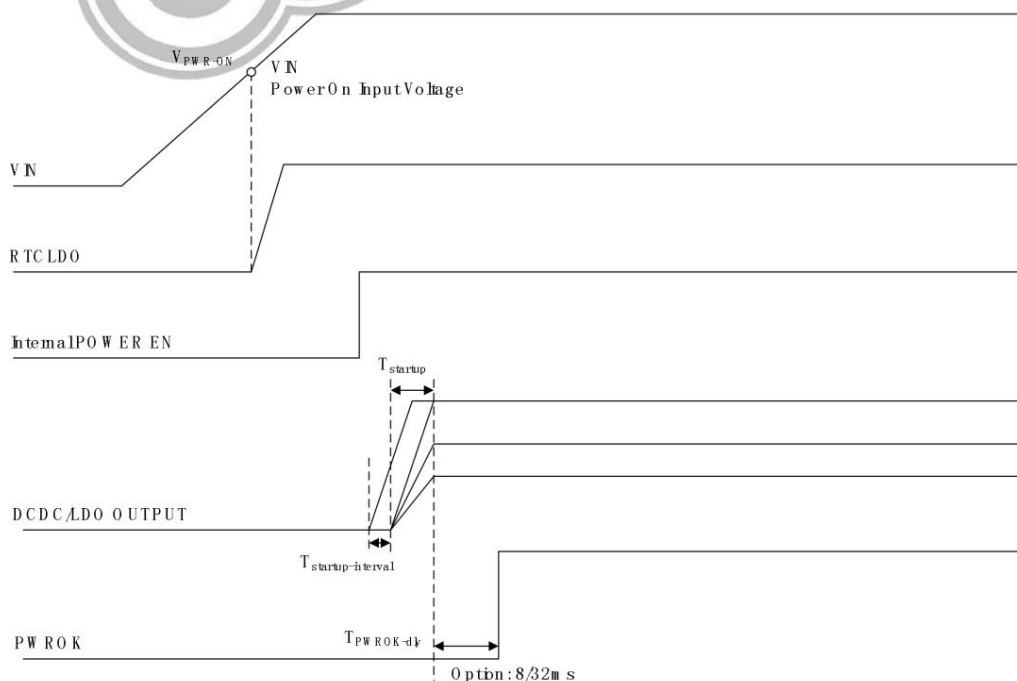
1. When EN/PWRON is used as PWRON, the boot sources include:

- (1) Power on by pressing the button. The POK button is pulled low for more than ONLEVEL.
- (2) VIN changes from low level to high level. This power-on function can be customized.
- (3) IRQ low level and internal 16ms debounce. This power-on function is customizable.

2. When EN/PWRON is used as EN, the chip powers on when the EN pin voltage changes from low to high (>0.6V).

Upon power-on, each power output starts according to a customized timing sequence.

Figure 6-2 Power-on sequence



Power off

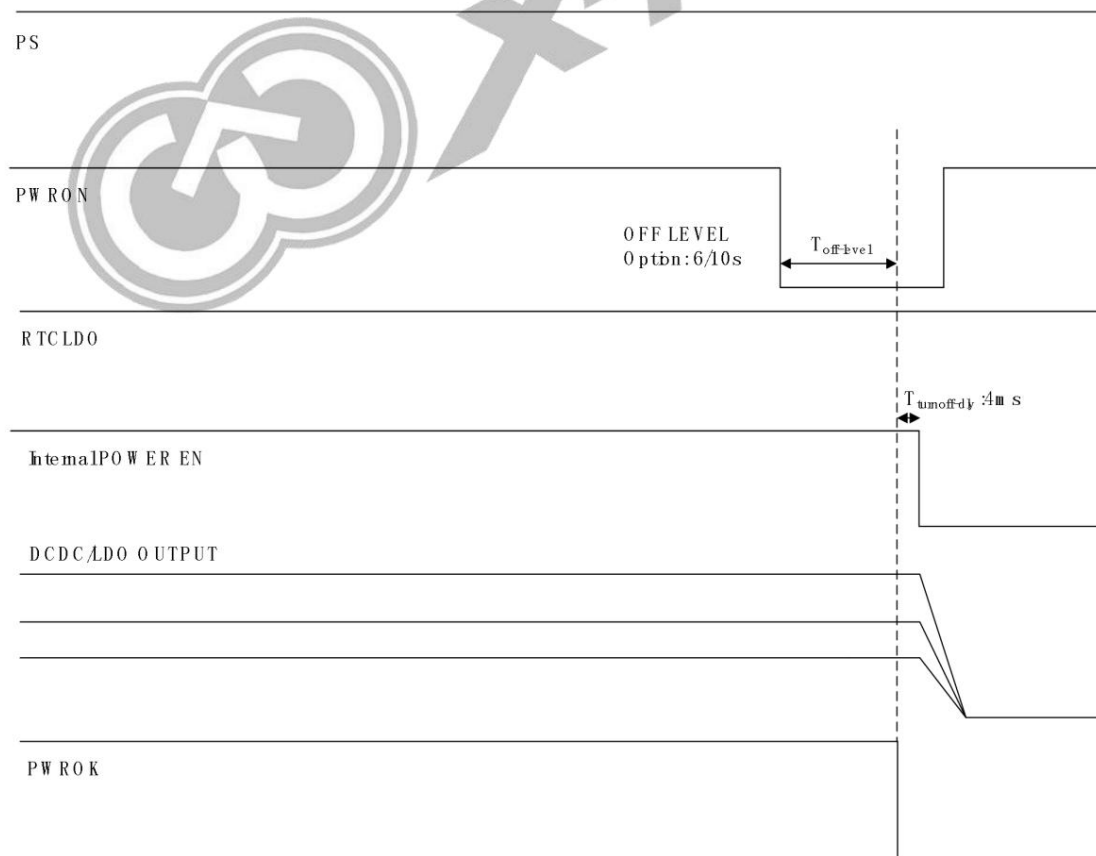
1. When EN/PWRON is used as PWRON, the power-off sources include:

- (1) Power off by button. The POK button is pulled low for more than OFFLEVEL. REG1BH[1] determines whether the power off function is enabled, and REG1BH[0] determines whether the power off function will restart after powering off.
- (2) Software shutdown. Write "1" to REG1AH[7].
- (3) VIN changes from high level to low level, i.e. $VIN < VOFF$ (default 2.6V).
- (4) $VIN > 5.8V$.
- (5) The DC-DC output voltage is lower than 85% of the set voltage. REG1DH[3:1] determines whether the shutdown function is enabled.
- (6) The DCDC output voltage is higher than 120% of the set voltage. REG1DH[0] determines whether the shutdown function is enabled.
- (7) DLDO1 has an overcurrent. REG1DH[4] determines whether the shutdown function is enabled.
- (8) The chip is overheating, exceeding the warning level (default 125 $^{\circ}$ C). REG1AH[1] determines whether the shutdown function is enabled.

2. When EN/PWRON is used as EN, the power-off sources include:

- (1) The voltage of the EN pin changes from high level to low level ($< 0.5V$).
- (2) Consistent with power-off source 3 to power-off source 8 used as PWRON pin.

Figure 6-3 Power-off sequence



Hibernation and Wake-up

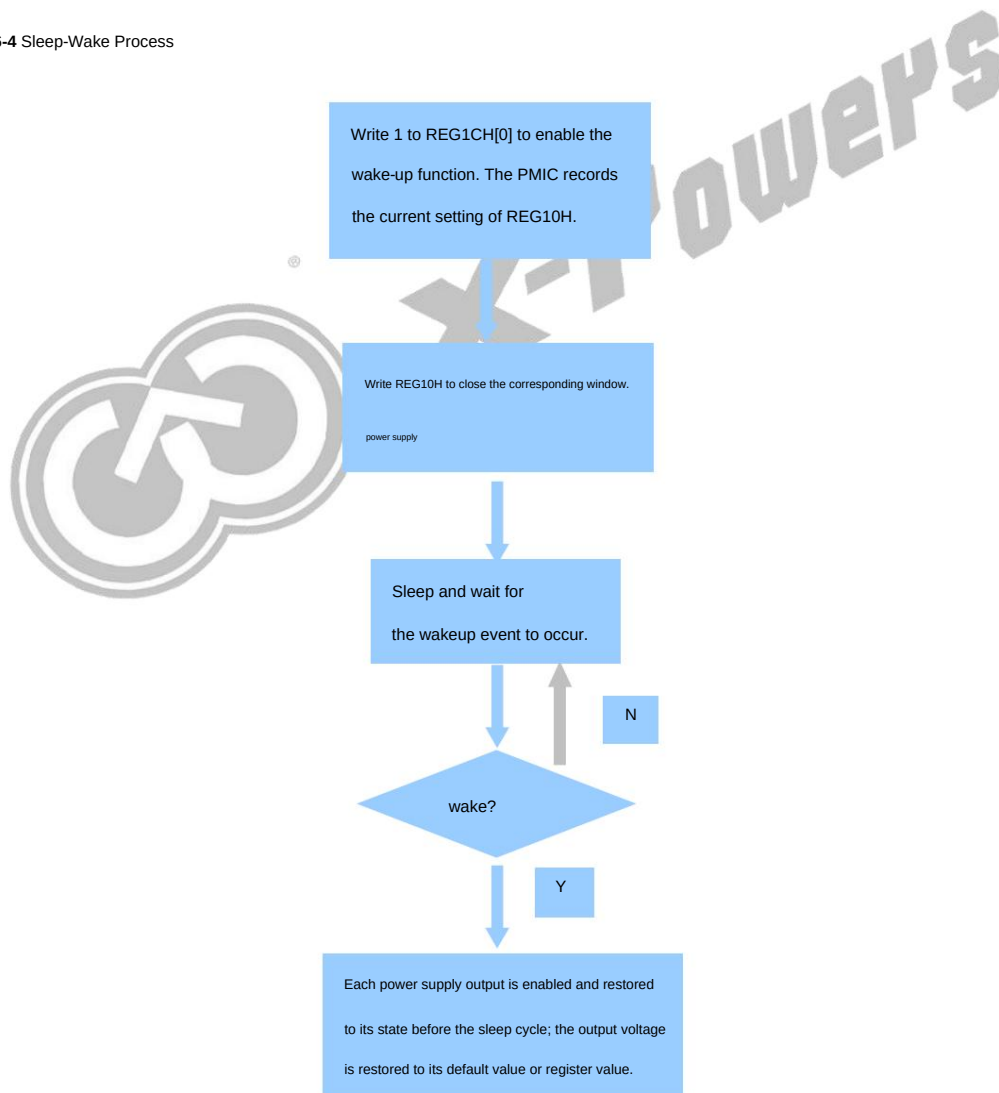
When the system is powered on, if it needs to enter Sleep mode and shut down one or more voltage outputs, REG1CH[0] can control whether wakeup is triggered by the following trigger signals:

1. Software wake-up. Write "1" to REG1CH[1].
2. IRQ pin low level wake-up. Write "1" to REG 1CH[4], and then pull the IRQ pin low by triggering the IRQ interrupt and debounce it for 16ms, i.e. (POKPIRQ(REG 20H[7]=1) or POKPNRQ(REG 20H[6]=1) or POKSIRQ(REG 20H[5]=1) or POKLIRQ(REG 20H[4]=1) or DCDC3 undervoltage interrupt(REG 20H[3]=1) or DCDC2 undervoltage interrupt(REG 20H[2]=1) or chip overtemperature interrupt(REG 20H[0]=1)); or externally pull the IRQ pin low and debounce it for 16ms.

These trigger sources cause the AXP313A to restore the state of each output power switch to the state before REG1CH[0] was written "1" and restore the voltage to the default value or register setting value (this function is determined by REG1CH[2]). Each power supply that was turned off is restored in sequence according to the set power-on sequence.

The following is the control flow for Sleep and Wakeup modes.

Figure 6-4 Sleep-Wake Process



Reset

The AXP313A has two reset methods: System reset and Power on reset (POR).

System reset

System reset refers to the PMIC shutting down and then powering on, i.e., performing a restart, where the power-down and power-on sequence of each power supply is the same as the normal power-on/off sequence.

The system will be consistent and the corresponding registers will be reset. During a system reset, RTCLDO remains on (if RTCLDO is configured to always be on).

A system reset can be achieved in the following two ways.

(1) With the device powered on, pull down PWROK.

The AXP313A's PWROK can be used as a reset signal for the application system. During power-on, PWROK outputs a low level, and when the power supply...

Once the output voltage stabilizes and reaches the set value, PWROK will be pulled high, thereby achieving a power-on reset of the application system.

If PWROK is pulled low via an external button or other means during normal operation of the application system, the AXP313A will shut down and then restart.

The power-on sequence of each output power supply is set according to the power-on sequence. This reset function is determined by REG1AH[4].

(2) Write "1" to REG1AH[6] to perform a software reset.

Power on reset (POR)

Power on reset (POR) means that all logic within the PMIC will undergo a reset operation. During POR, all DC-DC/LDOs, including...

RTCLDO will always power off first and then power on again.

6.5 Multiple power outputs

The following is a list of the multiple output voltages and functions provided by the AXP313A.

Table 6-1 Power Output Information

Output path type, default voltage, startup	steps, application recommendations		Maximum driving capability
DCDC1	BUCK	Customization Customization	GPU
DCDC2	BUCK		CPU
DCDC3	BUCK		DRAM
ALDO1	LDO		PLL/AVCC...
DLDO1	LDO		IO
RTCLDO	LDO		RTC

The AXP313A includes 3 synchronous buck DC-DC converters and 3 LDO converters. The DC-DC converters operate at 3MHz and can use small inductors and capacitors externally.

Capacitive components. All three DC-DC converters can be set to PWM mode or automatic mode (automatically switching according to the load size), see register REG12H.

All DC-DC/LDO converters support the discharge function, meaning that even when the output is off, the external output can be discharged through the internal discharge path.

The charge on the capacitor is released rapidly.

The default voltage and startup sequence of the DC-DC/LDO are customizable. There are four startup sequence levels (0-3), representing the default steps 1, 2, and 3 upon power-on.

Enables or disables this power supply. Additionally, the startup interval can be customized to 2/4/16/32ms.

6.6 Interruption

When certain specific events occur, the AXP313A alerts the host by pulling the IRQ low as an interrupt mechanism and saves the interrupt status.

In the register (see REG21H), writing a "1" to the corresponding status register bit clears the corresponding interrupt. When there are no interrupt events, the IRQ...

Pull high. Each interrupt can be masked via the interrupt control register (see REG22H).

Table 6-2 Interruption Information

Location	Interruption number	meaning
REG20H[7]	IRQ1	PWRON button rising edge
REG20H[6]	IRQ2	PWRON button falling edge
REG20H[5]	IRQ3	PWRON button short press
REG20H[4]	IRQ4	PWRON button long press
REG20H[3]	IRQ5	DCDC3 undervoltage
REG20H[2]	IRQ6	DCDC2 undervoltage
REG20H[0]	IRQ7	Chip overheating

6.7 Registers

6.7.1 Register List

address	describe	R/W	default value
00	Power-on source indicator register	R	
10	Power output switch control register	R/W	XXH
12	DC-DC PWM control register	R/W	00H
13	DCDC1 Voltage Setting Register	R/W	XXH
14	DCDC2 Voltage Setting Register	R/W	XXH
15	DCDC3 Voltage Setting Register	R/W	XXH
16	ALDO1 Voltage Setting Register	R/W	XXH
17	DLDO1 Voltage Setting Register	R/W	XXH
1A	Power off and restart control register	R/W	20H
1B	Power-off sequence, button power-off control register	R/W	06H
1C	Power Wake-up Control Register	R/W	00H
1D	Output monitoring and control register	R/W	1FH
1E	POK parameter setting register	R/W	80H
20	IRQ control register	R/W	31H
20H	IRQ Status Register	R/W	00H

6.7.2 Register Description

6.7.2.1 REG 00H: Power-on source indicator register

Reset: System reset

Bit	Description	R/W
7-5	Reserved	R
4	Power on from low to high when VIN is high, or power on from low to high when EN is high.	R
3	IRQ low level power-on	R
2	PWRON Button Power On	R
1	Reserved	R
0	VIN starts up from low to high	R

6.7.2.2 REG 10H: Power Output Switch Control Register

Default value: XXH

Reset: system reset

Bit	Description	R/W	default value
7-5	Reserved	RW	0
4	DLDO1 switch control	RW	custom made
3	ALDO1 switch control	RW	
2	DC-DC3 switch control	RW	
1	DC-DC2 switch control	RW	
0	DC-DC1 switch control	RW	

0: Off; 1: On

6.7.2.3 REG 12H: DC-DC PWM Control Register

Default value: 00H

Reset: system reset

Bit	Description	R/W	default value
7	DC-DC Spread Spectrum Function Settings 0: Off; 1: On	RW	0
6	DC-DC Spread Spectrum Frequency Settings 0: 50kHz; 1: 100kHz	RW	0
5	Retain	RW	0
4	Retain	RW	0
3	Retain	RW	0
2	DCDC-3 PFM/PWM Control 0: Automatic switching	RW	0

	1: Fixed PWM		
1	DCDC-2 PFM/PWM Control 0: Automatic switching 1: Fixed PWM	RW	0
0	DCDC-1 PFM/PWM Control 0: Automatic switching 1: Fixed PWM	RW	0

6.7.2.4 REG 13H: DC-DC1 Voltage Setting Register

Default value: XXH

Reset: system reset

Bit Description	R/W default value
7. Reserved	RW 0
6-0 DCDC1 voltage setting bit 6-0: 0.5~1.2V, 10mV/step, 71steps 1.22~1.54V, 20mV/step, 17steps 1.6~3.4V, 100mV/step, 19 steps	RW Custom

6.7.2.5 REG 14H: DC-DC2 voltage setting register

Default value: XXH

Reset: system reset

Bit Description	R/W default value
7. Reserved	RW 0
6-0 DCDC-2 voltage setting bit 6-0: 0.5~1.2V, 10mV/step, 71steps 1.22~1.54V, 20mV/step, 17steps	RW Custom

6.7.2.6 REG 15H: DC-DC3 voltage setting register

Default value: XXH

Reset: system reset

Bit Description	R/W default value
7. Reserved	RW 0
6-0 DCDC-3 voltage setting bit 6-0: 0.5~1.2V, 10mV/step, 71steps 1.22~1.84V, 20mV/step, 32steps	RW Custom

6.7.2.7 REG 16H: ALDO1 Voltage Setting Register

Default value: XXH

Reset: system reset

Bit Description	R/W	default value
7-5 Reserved	RW	0
4-0 ALDO1 Voltage setting bit 4-0: 0.5~3.5V, 100mV/step, 31steps	RW Custom	

6.7.2.8 REG 17H: DLDO1 Voltage Setting Register

Default value: XXH

Reset: system reset

Bit Description	R/W	default value
7-5 Reserved	RW	0
4-0 DLDO1 voltage setting bit4-0: 0.5~3.5V, 100mV/step, 31steps	RW Custom	

6.7.2.9 REG 1AH: Power-off/Reboot Control Register

Default value: 20H

Reset: bits [7:6] are for System reset, the other bits are for Power on reset.

Bit Description	R/W	default value
7. Software shutdown control. Writing 1 to this bit will turn off the PMIC output; this bit is automatically cleared to zero.	RW	0
6. Software restart control. Writing a 1 to this bit will restart the PMIC, and the bit will be automatically cleared. The		0
RW5 chip monitors the PWROK pin status during startup to determine if the power-on is normal. 0: No monitoring 1: Monitoring	RW	1
4 PWROK pulls low to restart PMIC function settings 0: Do not restart 1: Restart	RW	0
3-2 Retain	RW	0
1 PMIC internal over-temperature shutdown function settings 0: Do not turn off 1: Turn off	RW	0
0 Temperature protection threshold setting 0:125℃ 1:145℃	RW	0

6.7.2.10 REG 1BH: Power-off timing sequence, button power-off control register

Default value: 06H

Reset: Power on reset

Bit Description	R/W	default value
7-4 Reservation	RW	0000
3 Power output shutdown timing control 0: All outputs are turned off simultaneously. 1: Reverse of the boot sequence	RW	0
2. When powering off, after PWROK is pulled low, there is a 4ms delay before all outputs are turned off. 0: No delay 1: Delay	RW	1
1. Automatic shutdown function setting when button press time exceeds OFFLEVEL 0: Do not turn off 1: Turn off	RW	1
0. Should the device automatically restart after shutting down when the key press time exceeds OFFLEVEL? 0: Do not restart automatically 1: Restart automatically	RW	0

6.7.2.11 REG 1CH: Power-on Wake-up Control Register

Default value: 00H

Reset: bits [3] & [1:0] are for System reset, other bits are for Power on reset.

Bit Description	R/W	default value
7:5 Retained	RW	0
4 IRQ Pin Low-Level Wake-Up Function Setting 0: Close 1: Open	RW	0
3 Is PWROK pulled low during wake-up? 0: Do not lower 1: Lower	RW	0
2. Output voltage recovery control upon wake-up 0: Restore to default value 1: No voltage processing is performed (determined by the register setting).	RW	0
1. Software wake-up control. Writing 1 to this bit will restore all outputs; the bit will automatically clear to zero. RW		0
0 Sleep mode wake-up function enable settings 0: Wake-up function off 1: Wake-up function enabled	RW	0

6.7.2.12 REG 1DH: Output Monitoring and Control Register

Default value: 1FH

Reset: Power on reset

Bit Description	R/W	default value
7-5 Reserved	RW	000

4	DLDO1 Overcurrent Shutdown Function Settings 0: Off 1: On	RW	1
3	DCDC3 undervoltage shutdown function settings 0: Off 1: On	RW	1
2	DCDC2 undervoltage shutdown function settings 0: Off 1: On	RW	1
1	DCDC1 undervoltage shutdown function settings 0: Off 1: On	RW	1
0	DC-DC overvoltage shutdown function setting 0: Off 1: On	RW	1

6.7.2.13 REG 1EH: POK parameter setting register

Default value: 80H

Reset: Power on reset

Bit	Description	R/W	default value
7	Reserved	RW	1
6-2	Retain	RW	0
1	OFFLEVEL settings 0:6s 1:10s	RW	0
0	IRQLEVEL settings 0:1.5s 1:2.5s	RW	0

6.7.2.14 REG 20H: IRQ Control Register

Default value: 31H

Reset: System reset

Bit	Description	R/W	default value
7	PWRON rising edge IRQ enable	RW	0
6	PWRON Falling Edge IRQ Enable	RW	0
5	PWRON short press IRQ to enable	RW	1
4	PWRON: Press and hold IRQ to enable.	RW	1
3	DCDC3 Undervoltage IRQ Enable	RW	0
2	DCDC2 undervoltage IRQ enable	RW	0
1	Reserved	RW	0
0	Chip internal over-temperature IRQ enable	RW	1

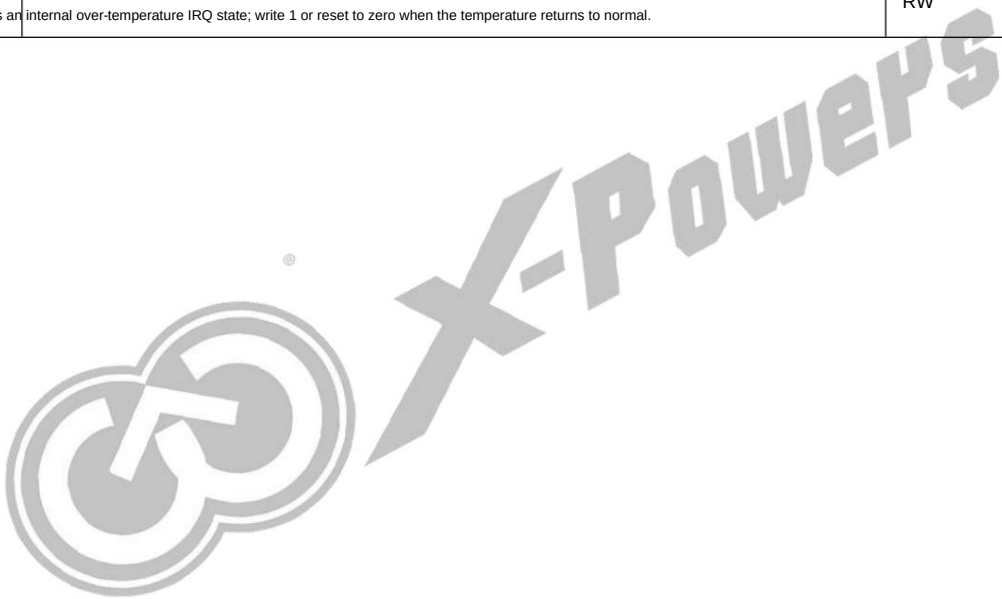


6.7.2.15 REG 21H: IRQ Status Register

Default value: 00H

Reset: System reset

Bit	Description	R/W	default value
7	PWRON rising edge IRQ state, write 1 to clear.	RW	0
6	PWRON falling edge IRQ state, write 1 to clear.	RW	0
5	In PWRON, a short press of the IRQ status bar resets the status to 1.	RW	0
4	In PWRON, long press the IRQ status button and write 1 to clear it.	RW	0
3	The DCDC3 is in an undervoltage IRQ state; write 1 or reset to zero when the voltage returns to normal.	RW	0
2	The DCDC2 is in an undervoltage IRQ state; write 1 or reset to zero when the voltage returns to normal.	RW	0
1	Reserved	RW	0
0	0 indicates an internal over-temperature IRQ state; write 1 or reset to zero when the temperature returns to normal.	RW	0



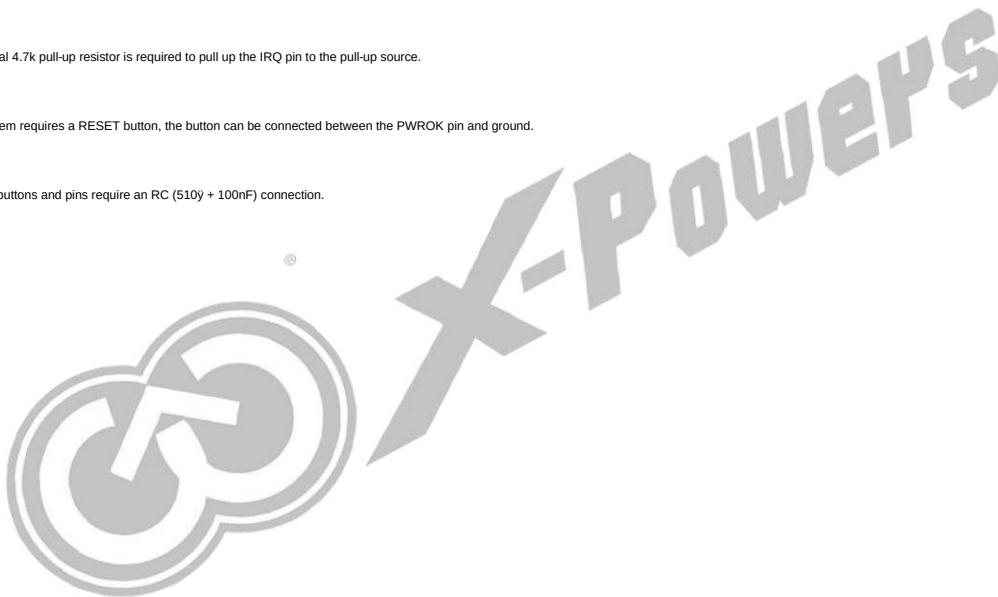
7. Application Information

7.1 DC-DC/LDO Design

- The output capacitance of the LDO is not less than 1uF.
- For LDOs that are customized to be unused (no output by default), simply leave their output pins floating; no capacitor is required.
- Each DC-DC converter should use a 1uH inductor with a saturation current greater than 30% of the load current and an internal resistance not exceeding 30mohm.
- The output capacitor of the DC-DC converter should be no less than 10uF.

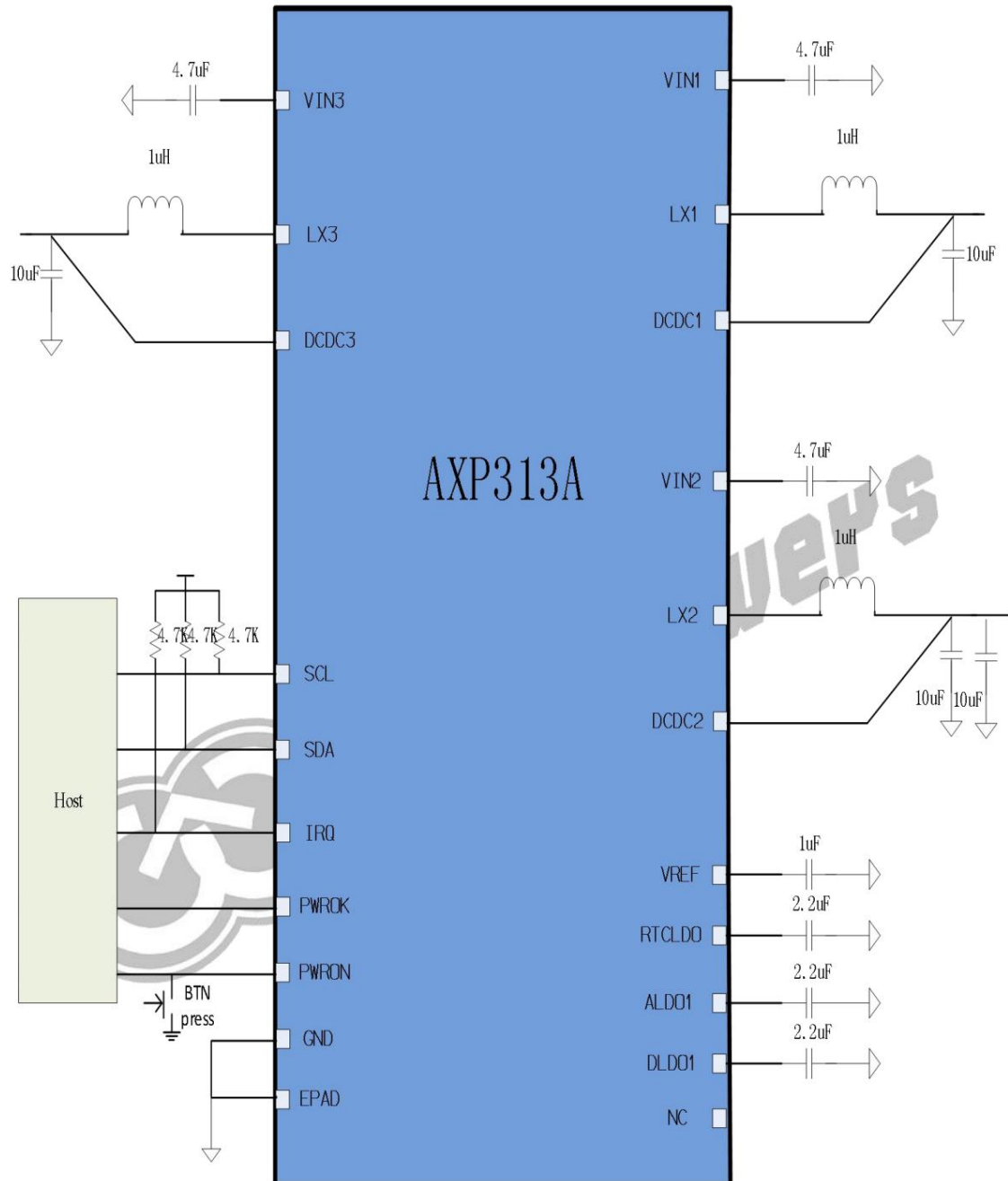
7.2 Design

- TWSS:SDA/SCL requires an external pull-up resistor to the pull-up source.
- An external 4.7k pull-up resistor is required to pull up the IRQ pin to the pull-up source.
- If the system requires a RESET button, the button can be connected between the PWROK pin and ground.
- PWROK buttons and pins require an RC (510 Ω + 100nF) connection.



7.3 Typical Applications

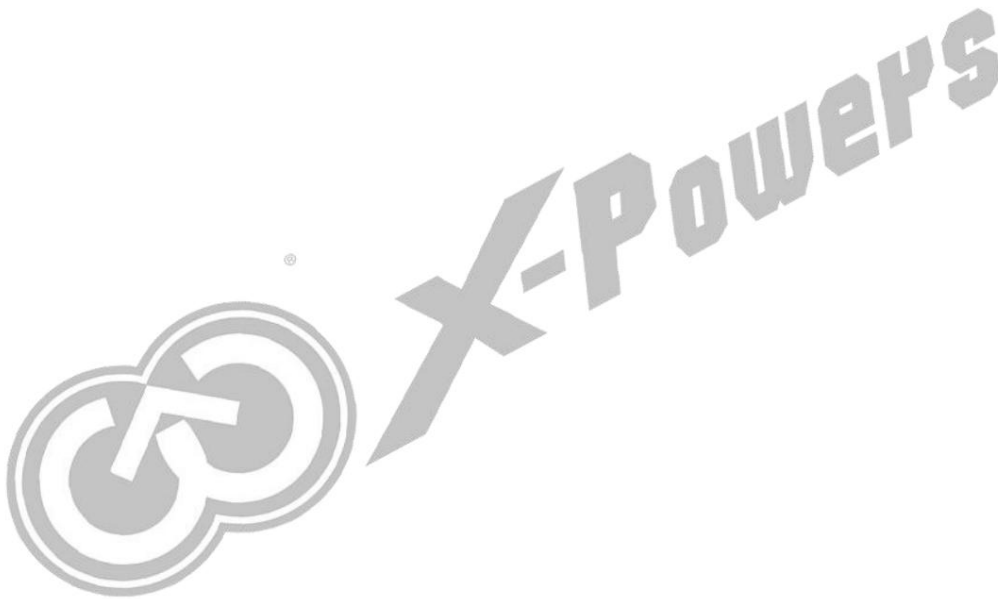
Figure 7-1 Typical Application Diagram



8 PCB Layout

High-current paths such as power input and output need to have wider traces to reduce trace impedance and thus reduce voltage drop and losses caused by the traces.

- DC-DC input and output linewidth: $\geq 150\text{mil}$.
- The LDO output line width is determined by the load current.
- The inductor of the DC-DC converter is close to the PMIC, the output capacitor is close to the inductor, and the input capacitor is close to the input pin of the PMIC.
- The DC-DC feedback line needs to be properly shielded to avoid the influence of high-frequency signals such as inductive switches.





9. Packaging, shipping, storage, and baking information

9.1 Packaging

The AXP313A is packaged in a QFN 3mm*3mm, 20-pin package. Figure 9-1 shows the AXP313A package.

Figure 9-1 Packaging Information

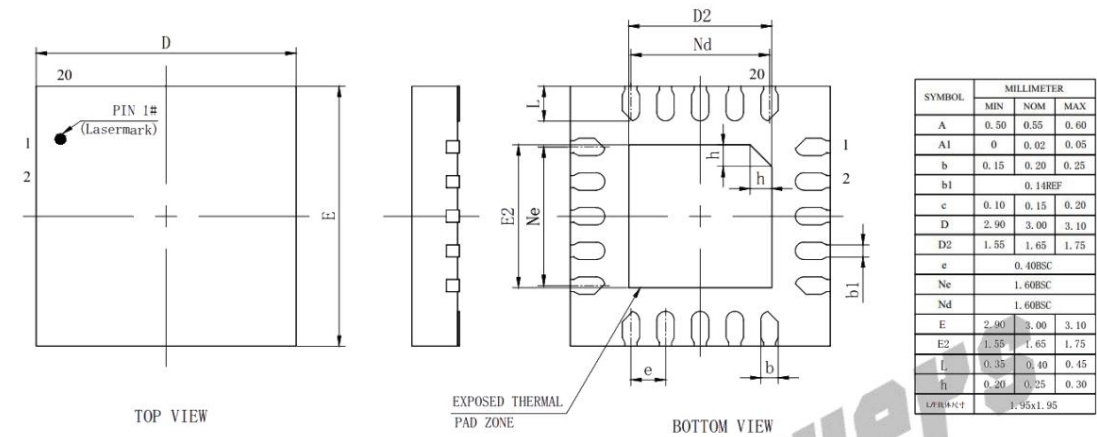


Figure 9-2 AXP313A silkscreen print

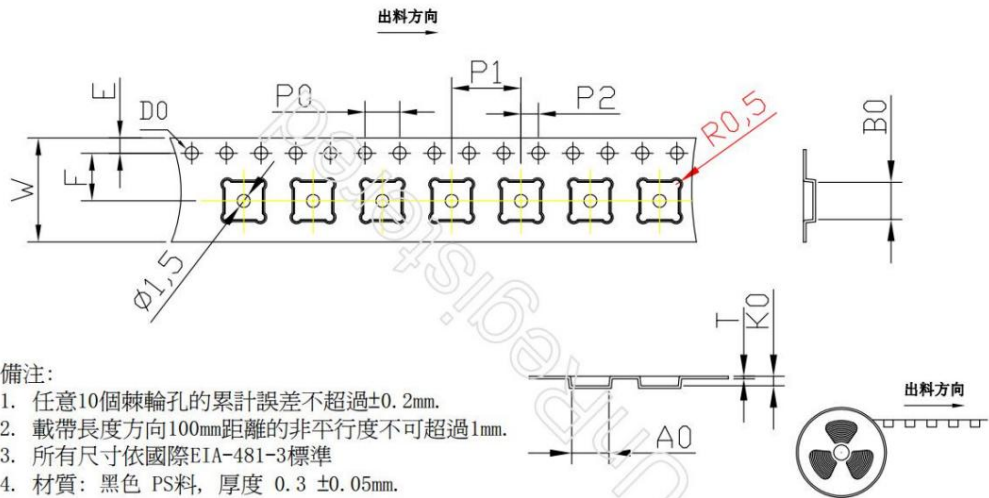


Table 9-1 AXP313A Silkscreen Information

No.	Marking	Description	Fixed/Dynamic
1	AXP313A	Product name	Fixed
2	LLLLLAA	Lot number	Dynamic
3	XXX1	Date code	Dynamic
4		X-POWERS logo	Fixed
5	White dot	Package pin 1	Fixed

9.2 Transportation

Figure 9-3 AXP313A tape and reel dimensions



ITEM	W	A0	B0	K0	P0	P1	P2	F	S	E	D0	T
DIM	12±0.30	3.30±0.1	3.30±0.1	1.10±0.05	4.0±0.1	8.00±0.1	2.0±0.1	5.5±0.1	0.0±0.1	1.75±0.1	1.5±0.05	0.3±0.05

Table 9-2 AXP313A Packaging Quantity Information

Type	Quantity	Part Number
Tape	3000pcs/Tape	AXP313A

9.3 Storage

9.3.1 Moisture Sensitivity Level (MSL)

The MSL (Mean Susceptibility Level) of a package indicates its ability to withstand exposure after being removed from the packaging. A low MSL means the component can be exposed to ambient moisture for a longer period of time than...

The higher the MSL, the longer the component. All MSL definitions are shown in Table 9-3.

Table 9-3 MSL Correspondence Table

MSL	Out-of-bag floor life	Comments
1	Unlimited	ÿ30ÿ/85%RH
2	1 year	ÿ30ÿ/60%RH
2a	4 weeks	ÿ30ÿ/60%RH
3	168 hours	ÿ30ÿ/60%RH
4	72 hours	ÿ30ÿ/60%RH
5	48 hours	ÿ30ÿ/60%RH
5a	24 hours	ÿ30ÿ/60%RH
6	Time on Label (TOL)	ÿ30ÿ/60%RH

The AXP313A is classified as MSL3.

9.3.2 Storage conditions inside the packaging

The storage life of AXP313A is defined in Table 9-4.

Table 9-4 Storage conditions inside the AXP313A package

Packing mode	Vacuum packing
Storage temperature	20~26
Storage humidity	40%~60%RH
Shelf life	6 months

9.3.3 Storage conditions outside packaging

According to the definition of MSL level, the external storage conditions of AXP313A are shown in Table 9-5.

Table 9-5 Storage conditions outside the packaging of AXP313A

Storage temperature	20~26
Storage humidity	40%~60%RH
Moisture Sensitivity Level(MSL)	3
Floor life	168 hours

For storage rules in this document, please refer to **IPC/ JEDEC J-STD-020C**.

9.4 Baking

If the storage conditions do not exceed those specified in Sections 9.4.2 and 9.4.3, the AXP313A does not need to be baked. If either condition is exceeded...

If the component is to be baked, then the AXP313A needs to be baked.

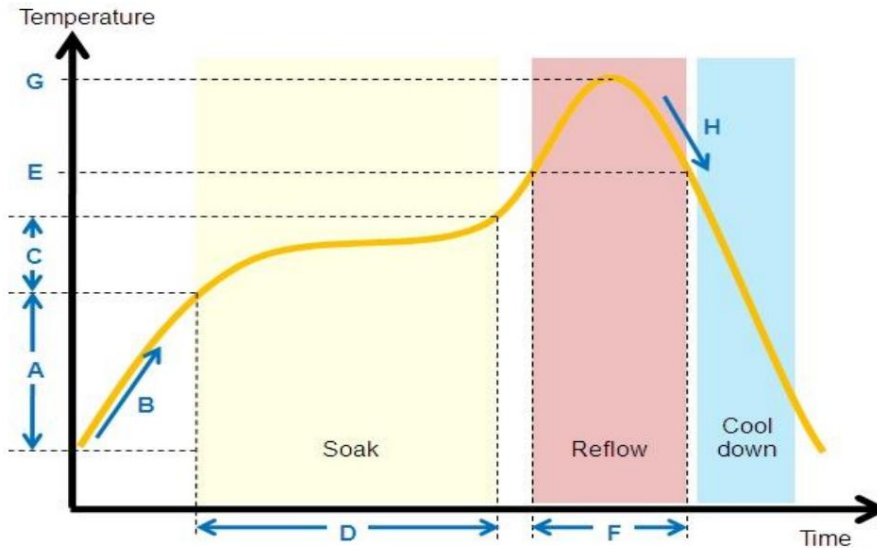
Table 9-6 Baking conditions for AXP313A

Surrounding	Condition	Note
Nitrogen	Tray: 125/8 hours	Recommended condition. It is Recommended to bake once, no More than three times.
	Tape: 60/72 hours	

10 Reflow Soldering Profile

The reflow soldering profile recommended in this article is a lead-free reflow soldering profile, suitable for pure lead-free technology using lead-free solder paste.

Figure 10-1 Typical reflow soldering profile of AXP313A



The reflow soldering profile conditions for AXP313A are shown in Table 10-1.

Table 10-1 AXP313A Reflow Soldering Profile Conditions

QTI typical SMT reflow profile conditions (for reference only)		
Step		Reflow condition
Environment	N2 purge reflow usage (yes/no)	Yes, N2 purge used
	If yes, O2 ppm level	O2 < 1500 ppm
A	Preheat ramp up temperature range 25 $\dot{\text{y}}$ -> 150 $\dot{\text{y}}$	
B	Preheat ramp up rate	1.5~2.5 $\dot{\text{y}}$ /sec
C	Soak temperature range	150 $\dot{\text{y}}$ -> 190 $\dot{\text{y}}$
D	Soak time	80~110 sec
E	Liquidus temperature	217 $\dot{\text{y}}$
F	Time above liquidus	60-90 sec
G	Peak temperature	240-250 $\dot{\text{y}}$
H	Cool down temperature rate	$\dot{\text{y}}$ 4 $\dot{\text{y}}$ /sec



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